

ENDG 319

Probability, Statistics and Machine Learning

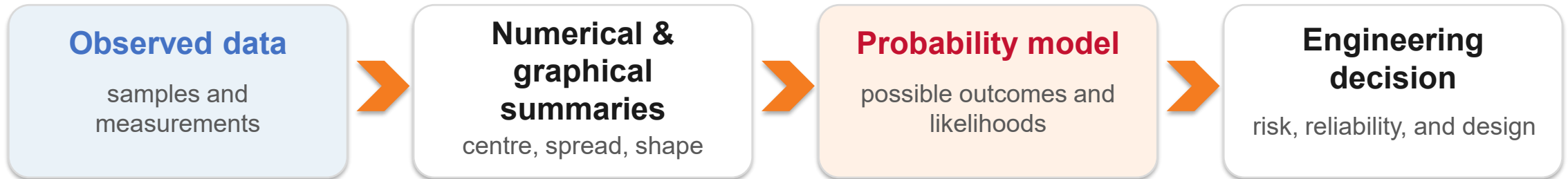


Lecture 4: Foundations of Probability

Hua Song, Ph.D., P.Eng.
Professor, Chemical & Petroleum Engineering
Schulich School of Engineering, University of Calgary
Fall 2026

From observed data to uncertain outcomes

Lectures 1–3 described data after they were collected. Probability models what may happen before the outcome is known.



Probability is the bridge from uncertainty to quantitative decisions.

Learning outcomes for Lecture 4

BY THE END OF TODAY, STUDENTS SHOULD BE ABLE TO

- Define a probability experiment, an outcome, and a sample space.
- Represent events as subsets of a sample space and determine when an event occurs.
- Use union, intersection, complement, and mutually exclusive events correctly.
- Interpret probability as long-run relative frequency and apply the probability axioms.
- Calculate probabilities using elementary outcomes, the addition rule, and the complement rule.

Core skill: translate an engineering uncertainty into a sample space, events, and probability statements.

Probability begins with an uncertain experiment

PROBABILITY EXPERIMENT

A process that produces an outcome that cannot be predicted with certainty in advance.

Manufacturing

Test a component:
defective or nondefective.

Measurement

Measure density or fill
volume.

Reliability

Operate a device: failure or
no failure.

Classical model

Roll a die or toss a coin.

The experiment is repeatable in principle; the particular outcome is uncertain.

Outcomes and the sample space

BASIC LANGUAGE

Outcome

The result obtained when the experiment is performed once.

Sample space S

The set of all possible outcomes of the experiment.

Example: roll a six-sided die

1

2

3

4

5

6

$$S = \{1, 2, 3, 4, 5, 6\}$$

After the roll, exactly one of these outcomes is observed.

Sample spaces may be discrete or continuous

SAMPLE SPACE

Discrete sample space

Outcomes can be listed or counted.

Coin toss: $S = \{H, T\}$

Die roll: $S = \{1, 2, 3, 4, 5, 6\}$

Component test: $S = \{D, N\}$

Continuous sample space

Outcomes form an interval or region.

For the measured volume x of a 330 mL container, under a stated measurement range:

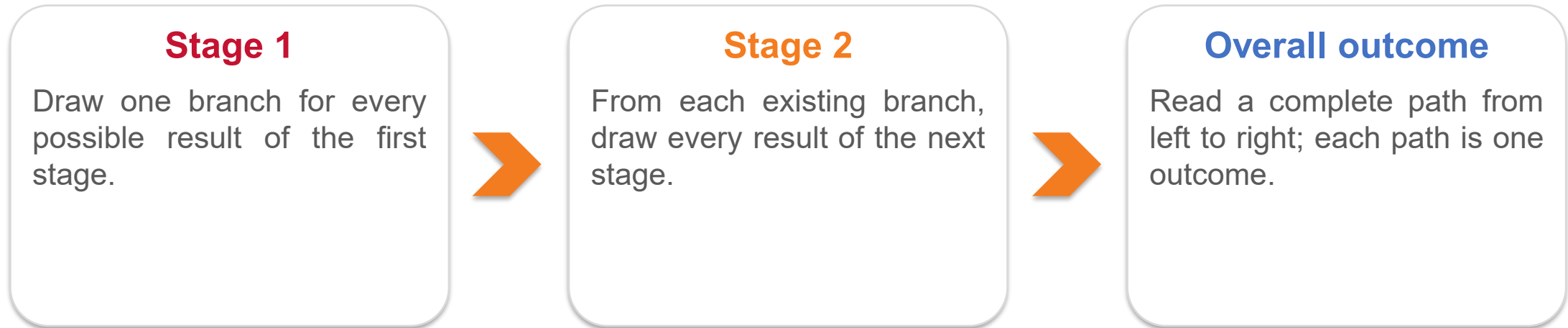
$$S = \{x : 310 < x < 350 \text{ mL}\}$$

A correct sample space must be exhaustive: every possible outcome belongs to S .

Enumerate compound outcomes systematically

TREE DIAGRAMS

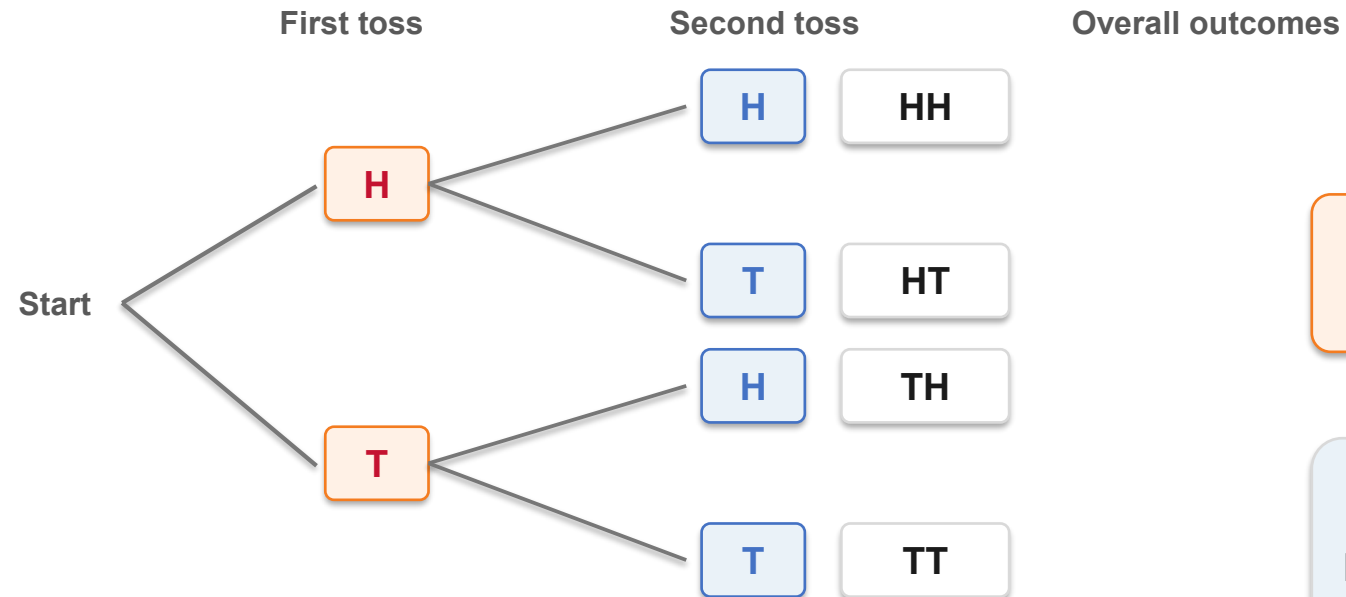
When an experiment has stages, a tree diagram keeps the sample space complete and avoids duplicate outcomes.



Product rule preview: if each of m stages has r possible results, the tree has r^m terminal paths.

Example: toss a coin twice

TREE DIAGRAM



$$S = \{HH, HT, TH, TT\}$$

Order matters

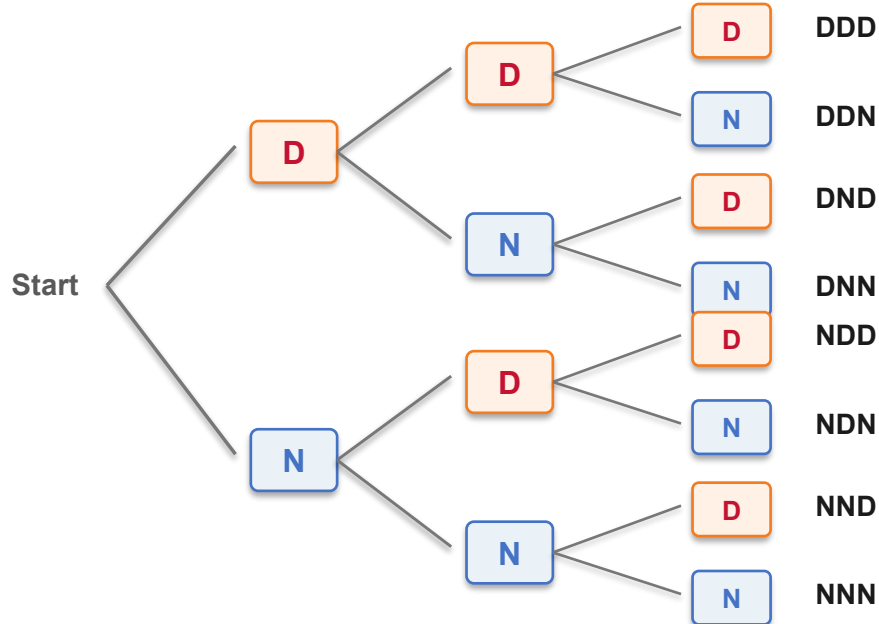
HT and TH are different outcomes because the sequence of tosses differs.

Check: does the tree contain every possible pair of results?

Example: test three items

TREE DIAGRAM

Each item is classified as defective (D) or nondefective (N).



Number of outcomes

$$2 \times 2 \times 2 = 8$$

$$S = \{DDD, DDN, DND, DNN, NDD, NDN, NND, NNN\}$$

Every terminal path records the classifications of items 1, 2, and 3.

An event is a subset of the sample space

EVENTS

Event $A \subseteq S$

An event collects the outcomes that satisfy a stated condition.

For one die roll: $S = \{1, 2, 3, 4, 5, 6\}$

A: an even number

$$A = \{2, 4, 6\}$$

B: a number > 3

$$B = \{4, 5, 6\}$$

Special events

Impossible: $\emptyset$
Certain: S

An event may contain one outcome, several outcomes, all outcomes, or no outcomes.

When does an event occur?

EVENT OCCURRENCE

An event occurs when the observed outcome belongs to that event.

Suppose the die is rolled and the observed outcome is 5.

Event	Did it occur?	Reason
$A = \{1, 2, 3\}$	No	$5 \notin A$
$B = \{4, 5, 6\}$	Yes	$5 \in B$
$C = \{3, 5, 6\}$	Yes	$5 \in C$
$D = \{6\}$	No	$5 \notin D$

Set operations describe combined events

PROBABILITY LANGUAGE

$$A \cup B$$

“A or B”

Outcomes in A, in B, or in both.

$$A \cap B$$

“A and B”

Outcomes common to both A
and B.

$$A^c$$

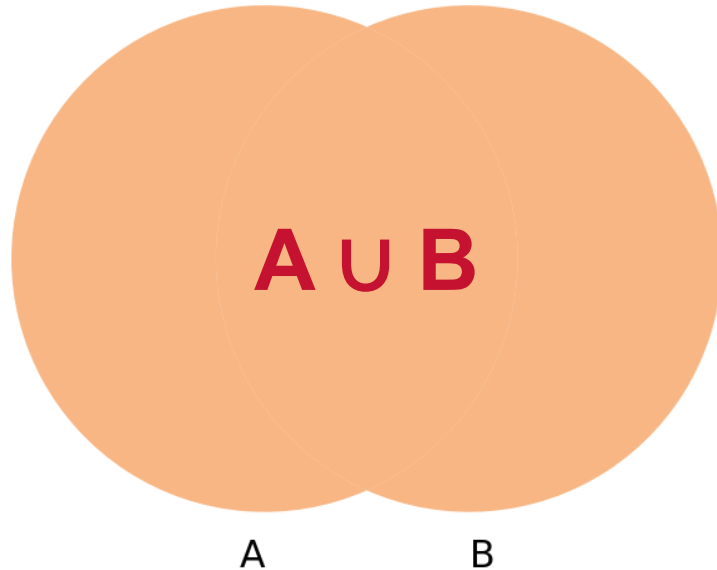
“not A”

Outcomes in S that are not in A.

Read the symbols in words before calculating: OR $\rightarrow$ union, AND $\rightarrow$ intersection, NOT $\rightarrow$ complement.

Union: at least one of the events occurs

$A \cup B$



For the die example:

$A = \{2, 4, 6\}$ (even)

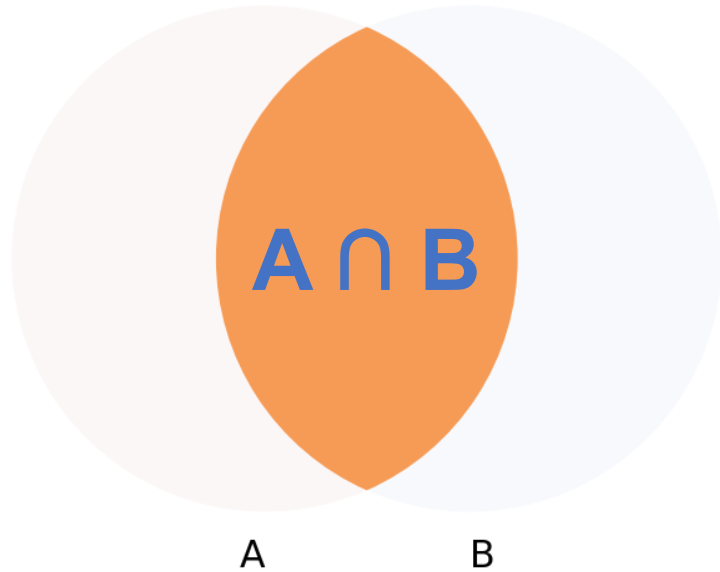
$B = \{4, 5, 6\}$ (> 3)

$$A \cup B = \{2, 4, 5, 6\}$$

The outcomes 4 and 6 belong to both sets, but each appears only once in the union.

Intersection: both events occur

$A \cap B$



For the same die events:

$A = \{2, 4, 6\}$ (even)

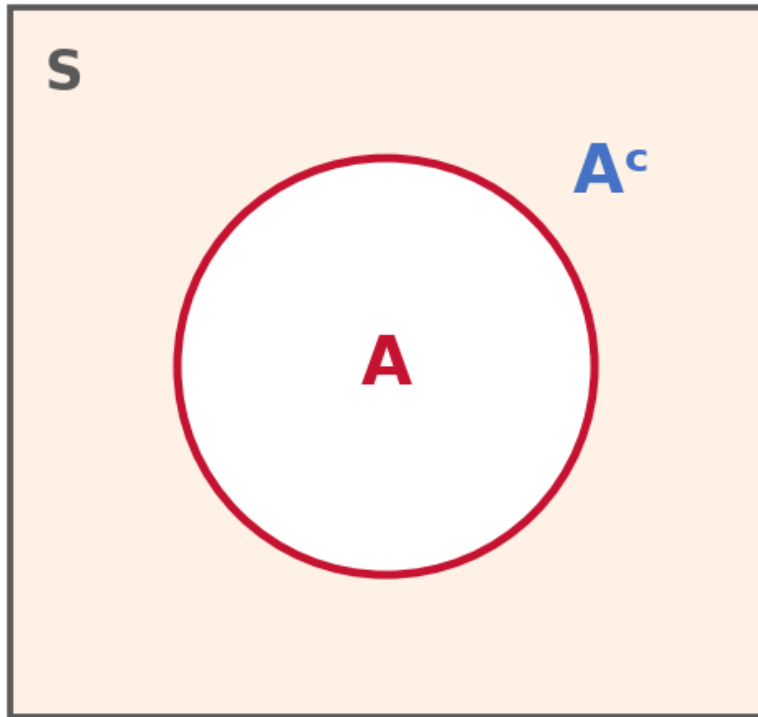
$B = \{4, 5, 6\}$ (> 3)

$$A \cap B = \{4, 6\}$$

Only outcomes that satisfy both descriptions belong to the intersection.

Complement: the event does not occur

A^c



For the die example:

$S = \{1, 2, 3, 4, 5, 6\}$

$A = \{2, 4, 6\}$ (even)

$$A^c = \{1, 3, 5\}$$

A and A^c are disjoint and together cover the entire sample space.

Mutually exclusive events have no outcomes in common

DISJOINT EVENTS

$$A \cap B = \emptyset$$

The events cannot occur simultaneously in one performance of the experiment.

Mutually exclusive

$A = \{2, 4, 6\}$ (even)

$B = \{1, 3, 5\}$ (odd)

$$A \cap B = \emptyset$$

Not mutually exclusive

$A = \{2, 4, 6\}$ (even)

$C = \{4, 5, 6\}$ (> 3)

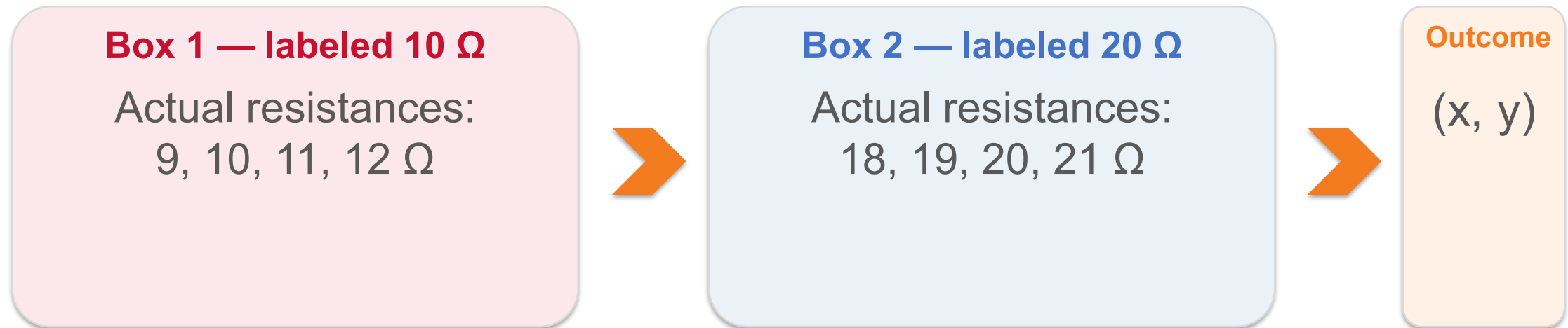
$$A \cap C = \{4, 6\}$$

Mutual exclusivity is determined by the sets—not by whether events merely sound different.

Engineering example: selecting two resistors

WORKED EXAMPLE

An electrical engineer selects one resistor from each box and measures the two resistances.



Because one value is selected from each box, each outcome is an ordered pair.

Resistor experiment: construct the sample space

WORKED EXAMPLE

Four choices from Box 1 $\times$ four choices from Box 2 = 16 possible ordered pairs.

Box 1 ↓ Box 2 →	18	19	20	21
9	(9, 18)	(9, 19)	(9, 20)	(9, 21)
10	(10, 18)	(10, 19)	(10, 20)	(10, 21)
11	(11, 18)	(11, 19)	(11, 20)	(11, 21)
12	(12, 18)	(12, 19)	(12, 20)	(12, 21)

S contains every pair (x, y) that can be produced by the selection procedure.

Resistor experiment: identify the events

WORKED EXAMPLE

Box 1 ↓ Box 2 →	18	19	20	21
9	(9, 18)	(9, 19)	(9, 20)	(9, 21)
10	(10, 18)	(10, 19)	(10, 20)	(10, 21)
11	(11, 18)	(11, 19)	(11, 20)	(11, 21)
12	(12, 18)	(12, 19)	(12, 20)	(12, 21)

Event A

First resistance $> 10 \Omega \rightarrow x \in \{11, 12\}$

Event B

Second resistance $< 19 \Omega \rightarrow y = 18$

Event C

Sum of resistances $= 28 \Omega \rightarrow (9, 19), (10, 18)$

Shading is a visual aid; the event itself is the set of ordered pairs satisfying the condition.

Resistor experiment: which events can occur together?

WORKED EXAMPLE

$$A \cap B$$

$$\{(11,18), (12,18)\}$$

Both can occur.

$$B \cap C$$

$$\{(10,18)\}$$

Both can occur.

$$A \cap C$$

$$\emptyset$$

A and C are mutually
exclusive.

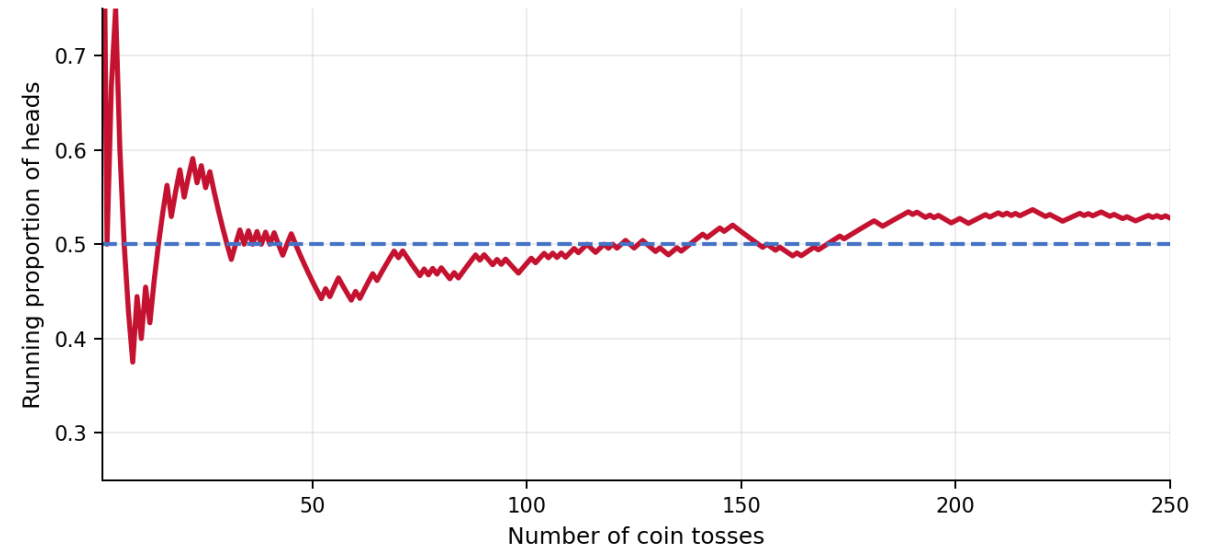
Method: translate “both” into an intersection and look for common outcomes.

Probability quantifies long-run likelihood

INTERPRETATION

$P(A)$ denotes the probability that event A occurs.

If an experiment is repeated many times under the same conditions, the relative frequency of A tends to stabilize near $P(A)$.



Relative frequency of A = number of times A occurs / number of repetitions

Probability is a model parameter; an observed relative frequency is a data-based estimate.

The axioms of probability

FOUNDATION

1 Bounds

For every event A ,
 $0 \leq P(A) \leq 1$

2 Certainty

For the sample space S ,
 $P(S) = 1$

3 Additivity

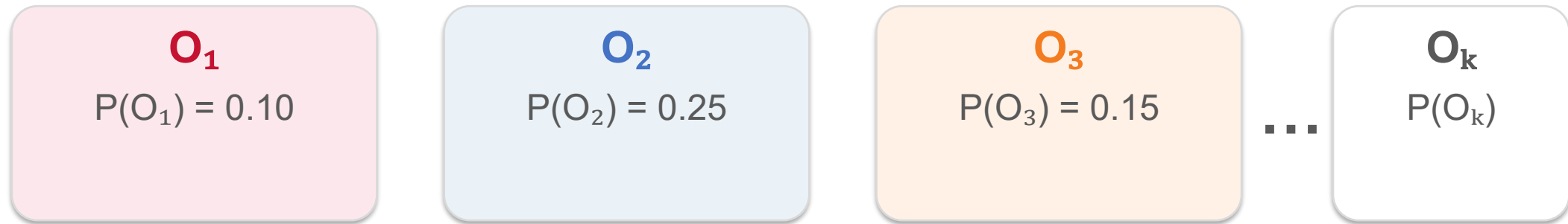
If A and B are mutually
exclusive,
 $P(A \cup B) = P(A) + P(B)$

These axioms are enough to derive the complement rule and the general addition rule.

Find an event probability by adding outcome probabilities

ELEMENTARY OUTCOMES

Suppose event A contains the mutually exclusive elementary outcomes $O_1, O_2, \dots, O_k$.



$$P(A) = P(O_1) + P(O_2) + \dots + P(O_k)$$

Why can we add? Distinct elementary outcomes cannot occur simultaneously.

Equally likely outcomes: count favorable outcomes carefully

CLASSICAL PROBABILITY

$$P(A) = k / N$$

N

Number of outcomes in S

k

Number of outcomes in A

Derivation: If N outcomes are equally likely, each has probability $1/N$.
If A contains k of those outcomes, add $1/N$ exactly k times.

Critical condition: “favorable outcomes / total outcomes” is valid only when the outcomes are equally likely.

Quick example: a fair die is rolled

EQUALLY LIKELY OUTCOMES

What is the probability that an even number occurs?



S has $N = 6$ equally likely outcomes.

$A = \{2, 4, 6\}$ has $k = 3$ outcomes.

$$P(A) = 3 / 6 = 1 / 2 = 0.50$$

Addition rule for mutually exclusive events

SPECIAL CASE

If $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$.

Example: one die roll

$$A = \{1, 2\} \quad B = \{5, 6\}$$
$$A \cap B = \emptyset$$

$$P(A \cup B) = 2/6 + 2/6 = 4/6$$

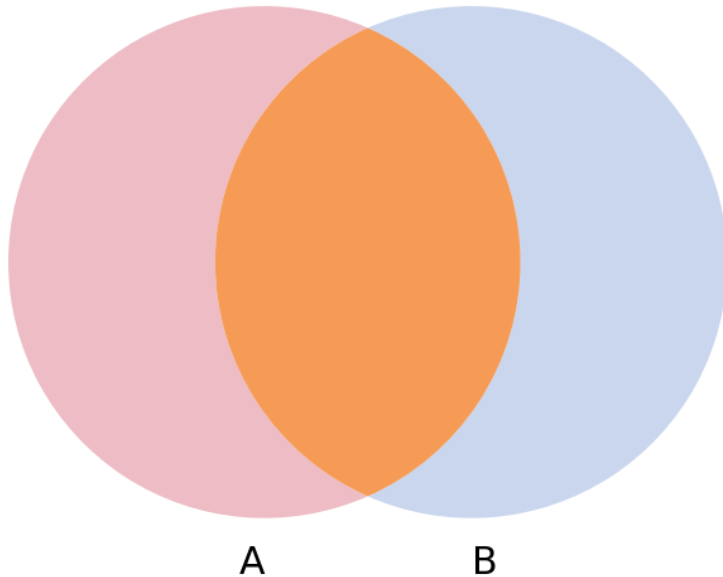
Why direct addition works

No outcome belongs to both events, so no probability is counted twice.

Before adding probabilities, always check whether the events overlap.

General addition rule: subtract the overlap once

A OR B



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

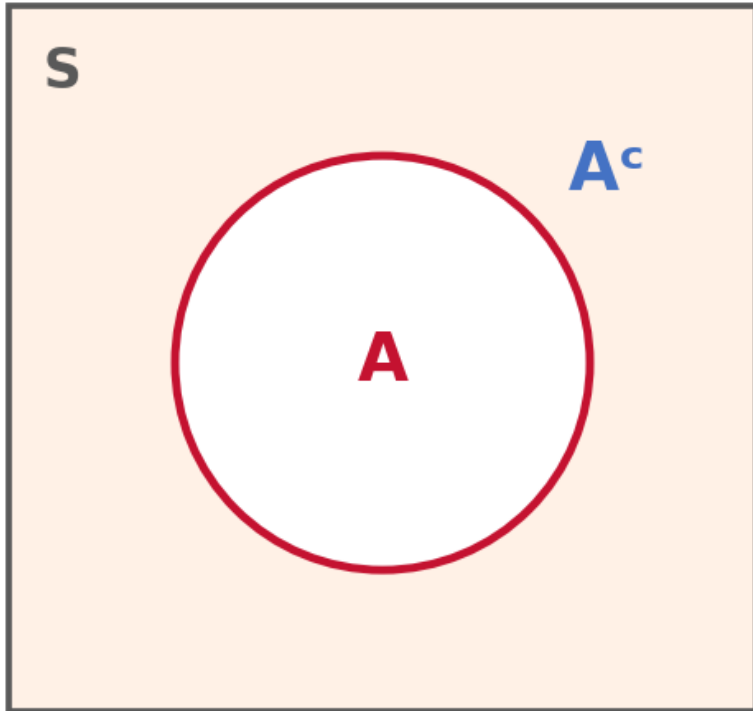
Why subtract?

- $P(A)$ counts the overlap once.
- $P(B)$ counts the same overlap again.
- Subtract $P(A \cap B)$ to remove one copy.

When A and B are mutually exclusive, $P(A \cap B) = 0$ and the formula reduces to direct addition.

Rule of complement

NOT A



Because A and A^c are mutually exclusive and exhaustive:

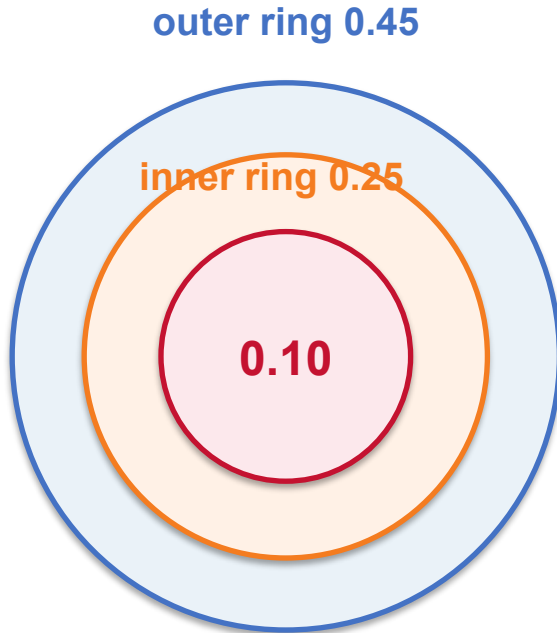
$$P(A) + P(A^c) = P(S) = 1$$

$$P(A^c) = 1 - P(A)$$

The complement is often the shortest route for “at least one,” “none,” or “not” questions.

Example: probability of hitting a target

MUTUALLY EXCLUSIVE REGIONS



The three hit regions do not overlap.

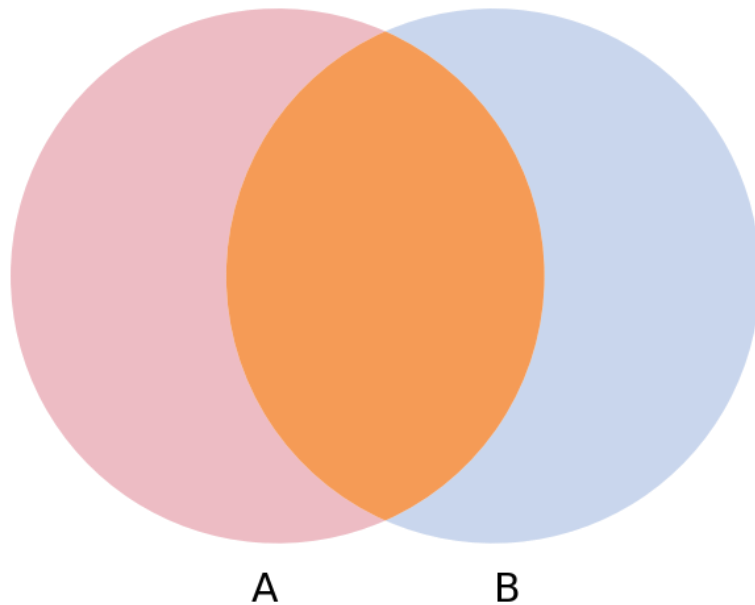
$$P(\text{hit}) = 0.10 + 0.25 + 0.45 = 0.80$$

$$P(\text{miss}) = 1 - P(\text{hit}) = 0.20$$

Use direct addition for disjoint regions; use the complement for the miss probability.

Example: an aluminum can with overlapping flaws

GENERAL ADDITION RULE



S: side flaw **T: top flaw**

$$P(S) = 0.02 \quad P(T) = 0.03 \quad P(S \cap T) = 0.01$$

$$P(\text{any flaw}) = 0.02 + 0.03 - 0.01 = 0.04$$

$$P(\text{no flaw}) = 1 - 0.04 = 0.96$$

The “both” category is subtracted because it is included in each individual flaw probability.

In-class concept check

- 1 For two coin tosses, write the event “exactly one head.”
- 2 If $A = \{1,2,3\}$ and $B = \{3,4\}$, are A and B mutually exclusive?
- 3 If $P(A) = 0.62$, find $P(A^c)$.
- 4 If $P(A)=0.40$, $P(B)=0.35$, and $P(A \cap B)=0.10$, find $P(A \cup B)$.

Key takeaways

- A probability experiment produces an uncertain outcome; the sample space S contains all possible outcomes.
- An event is a subset of S and occurs when the observed outcome belongs to that subset.
- Union means OR, intersection means AND, and complement means NOT.
- Mutually exclusive events have empty intersection and cannot occur simultaneously.
- Probability satisfies $0 \leq P(A) \leq 1$, $P(S)=1$, and additivity for mutually exclusive events.
- Use k/N only for equally likely outcomes; use the general addition and complement rules when appropriate.

Next: probability rules for sequential and dependent events.